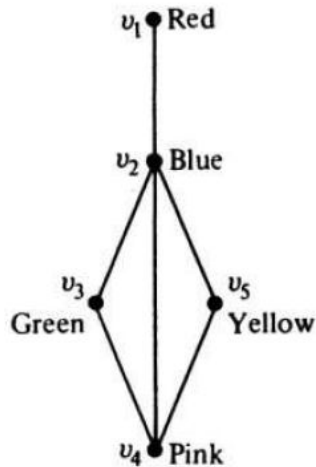


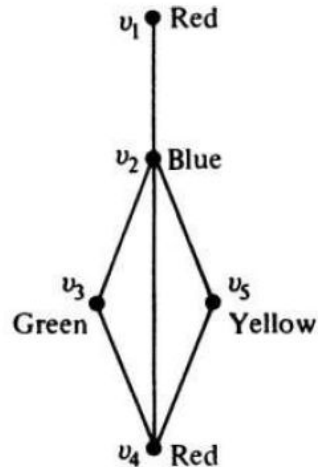
Graph Coloring, Covering and Partitioning

Chromatic Number

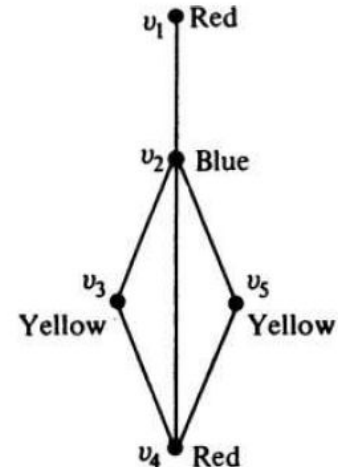
Painting all the vertices of a graph with colors such that no two adjacent vertices have the same color is called the **proper coloring** (or sometimes simply coloring) of a graph.



(a)



(b)



(c)

Chromatic Number

The proper coloring which is of interest to us is one that requires the minimum number of colors.

A graph G that requires κ different colors for its proper coloring, and no less, is called a κ -chromatic graph, and the number κ is called the chromatic number of G .

You can verify that the graph in the previous slide is 3-chromatic.

Chromatic Number

Some observations:

1. A graph consisting of only isolated vertices is 1-chromatic.
2. A graph with one or more edges (not a self-loop, of course) is at least 2-chromatic (also called bichromatic).
3. A complete graph of n vertices is n -chromatic, as all its vertices are adjacent. Hence a graph containing a complete graph of r vertices is at least r -chromatic. For instance, every graph having a triangle is at least 3-chromatic.

Chromatic Number

Some observations:

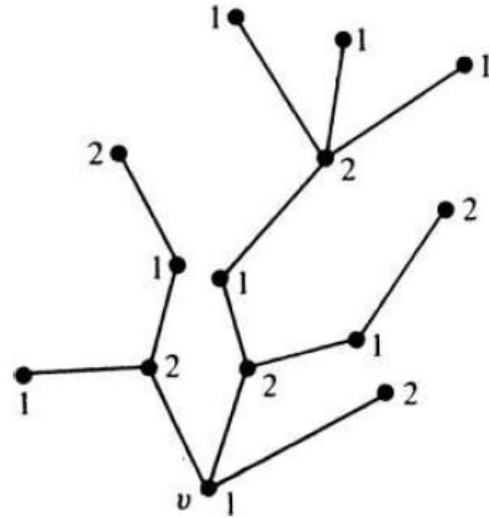
4. A graph consisting of simply one circuit with $n \geq 3$ vertices is 2-chromatic if n is even and 3-chromatic if n is odd. (This can be seen by numbering vertices $1, 2, \dots, n$ in sequence and assigning one color to odd vertices and another to even. If n is even, no adjacent vertices will have the same color. If n is odd, the n th and first vertex will be adjacent and will have the same color, thus requiring a third color for proper coloring.)

THEOREM 8-1

“Every tree with two or more vertices is 2-chromatic.”

Select any vertex v in the given tree T . Consider T as a rooted tree at vertex v . Paint v with color 1. Paint all vertices adjacent to v with color 2. Next, paint the vertices adjacent to these (those that just have been colored with 2) using color 1. Continue this process till every vertex in T has been painted.

Now in T we find that all vertices at odd distances from v have color 2, while v and vertices at even distances from v have color 1.

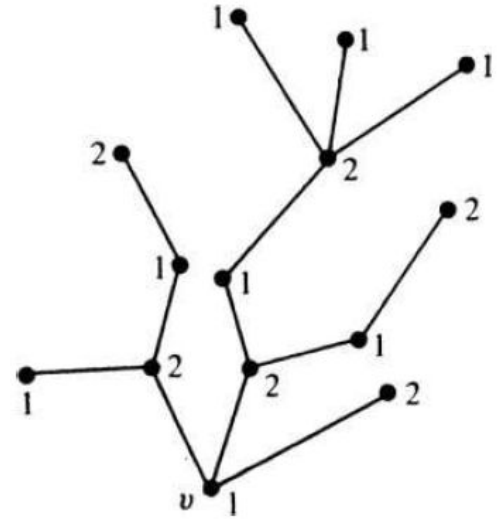


THEOREM 8-1

“Every tree with two or more vertices is 2-chromatic.”

Now along any path in T the vertices are of alternating colors. Since there is one and only one path between any two vertices in a tree, no two adjacent vertices have the same color. Thus T has been properly colored with two colors.

Though a tree is 2-chromatic, not every 2-chromatic graph is a tree. Theorem 8-2 will characterize all 2-chromatic graphs.



THEOREM 8-2

“A graph with at least one edge is 2-chromatic if and only if it has no circuits of odd length.”

Let G be a connected graph with circuits of only even lengths. Consider a spanning tree T in G . Using the coloring procedure and the result of Theorem 8-1, let us properly color T with two colors.

Now add the chords to T one by one. Since G had no circuits of odd length, the end vertices of every chord being replaced are differently colored in T .

Thus G is colored with two colors, with no adjacent vertices having the same color. That is, G is 2-chromatic.

THEOREM 8-3

If $d_{\max}$ is the maximum degree of the vertices in a graph G ,

$$\text{chromatic number of } G \leq 1 + d_{\max}$$

This upper bound can be improved by 1 if G has no complete graph of $d_{\max} + 1$ vertices. In that case

$$\text{chromatic number of } G \leq d_{\max}$$

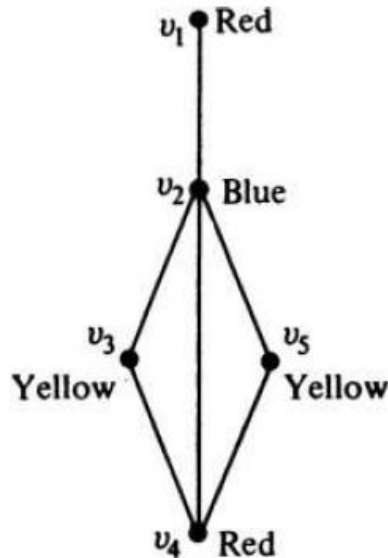
Bipartite Graphs

A graph G is called bipartite if its vertex set V can be decomposed into two disjoint subsets V_1 and V_2 such that every edge in G joins a vertex in V_1 with a vertex in V_2 . Thus every tree is a bipartite graph.

Clearly, every 2-chromatic graph is bipartite because the coloring partitions the vertex set into two subsets V_1 and V_2 such that no two vertices in V_1 (or V_2) are adjacent. Similarly, every bipartite graph is 2-chromatic, with one trivial exception; a graph of two or more isolated vertices and with no edges is bipartite but is 1-chromatic.

CHROMATIC PARTITIONING

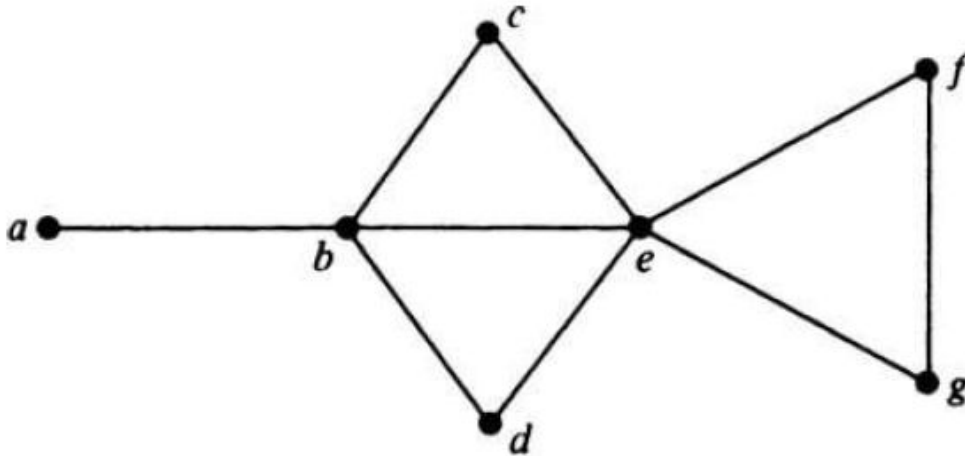
A proper coloring of a graph naturally induces a partitioning of the vertices into different subsets. For example, the coloring in the following graph produces the partitioning $\{v_1, v_4\}$, $\{v_2\}$, and $\{v_3, v_5\}$.



CHROMATIC PARTITIONING

A set of vertices in a graph is said to be an independent set of vertices or simply an independent set if no two vertices in the set are adjacent.

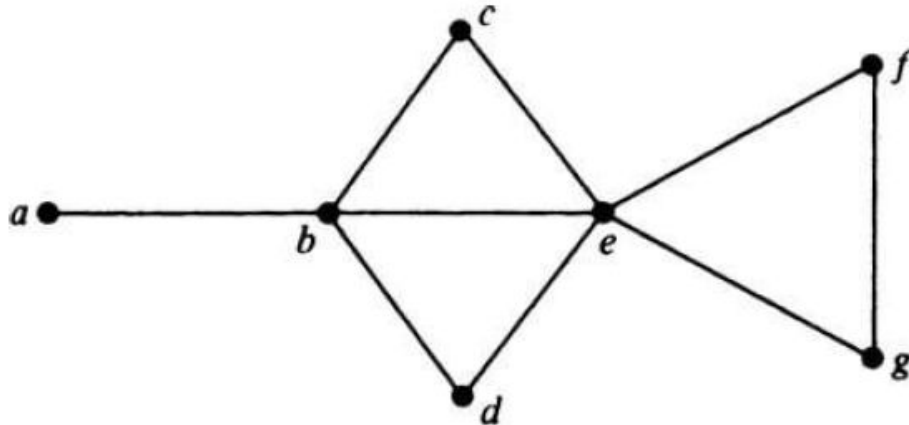
For example, in the graph shown here, $\{a, c, d\}$ is an independent set.



CHROMATIC PARTITIONING

Obviously, a single vertex in any graph constitutes an independent set. But we are mainly interested in the **maximal independent set**, which is such an independent set to which no other vertex can be added without destroying its independence property.

The set $\{a, c, d, f\}$ is a maximal independent set. The set $\{b, f\}$ is another maximal independent set. The set $\{b, g\}$ is a third one.



CHROMATIC PARTITIONING

The number of vertices in the largest independent set of a graph G is called the independence number (or coefficient of internal stability), $\beta(G)$.

Consider a κ -chromatic graph G of n vertices properly colored with κ different colors. Since the largest number of vertices in G with the same color cannot exceed the independence number $\beta(G)$, we have the inequality

$$\beta(G) \geq \frac{n}{\kappa}.$$

Maximal Independent Set

A reasonable method of finding a maximal independent set in a graph G will be to start with any vertex v of G in the set. Add more vertices to the set, selecting at each stage a vertex that is not adjacent to any of those already selected.

This procedure will ultimately produce a maximal independent set. This set, however, is not necessarily a maximal independent set with a largest number of vertices.

Maximal Independent Set

A reasonable (but not very efficient for large graphs) method for obtaining all maximal independent sets in any graph can be developed using Boolean arithmetic on the vertices.

Let each vertex in the graph be treated as a Boolean variable. Let the logical (or Boolean) sum $a + b$ denote the operation of including vertex a or b or both; let the logical multiplication ab denote the operation of including both vertices a and b , and let the Boolean complement a' denote that vertex a is not included.

Maximal Independent Set

For a given graph G we must find a maximal subset of vertices that does not include the two end vertices of any edge in G . Let us express an edge (x, y) as a Boolean product, xy , of its end vertices x and y , and let us sum all such products in G to get a Boolean expression

$$\varphi = \sum xy \text{ for all } (x, y) \text{ in } G$$

Let us further take the Boolean complement φ' of this expression, and express it as a sum of Boolean products

$$\varphi' = f_1 + f_2 + \dots + f_k$$

Maximal Independent Set

A vertex set is a maximal independent set if and only if $\varphi = 0$ (logically false), which is possible if and only if $\varphi' = 1$ (true), which is possible if and only if at least one $f_i = 1$, which is possible if and only if each vertex appearing in f_i , (in complemented form) is excluded from the vertex set of G .

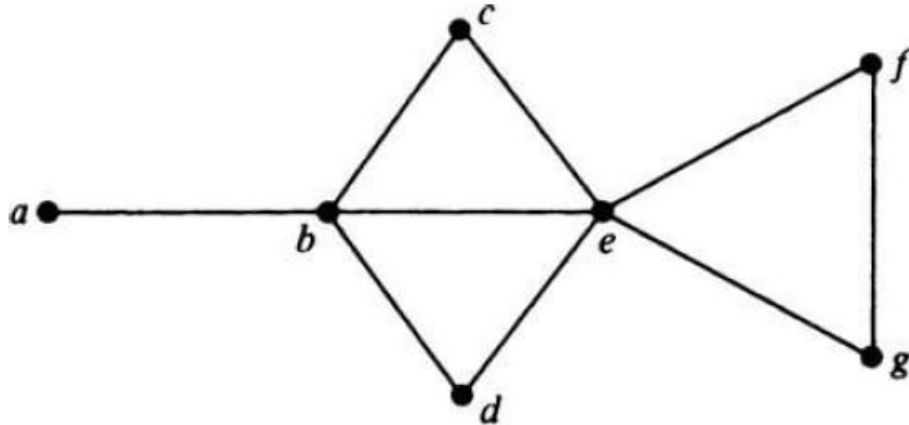
Thus each f_i will yield a maximal independent set, and every maximal independent set will be produced by this method.

Maximal Independent Set - Example

For the previous graph,

$$\varphi = ab + bc + bd + be + ce + de + ef + eg + fg,$$

$$\varphi' = (a' + b')(b' + c')(b' + d')(b' + e')(c' + e')(d' + e')(e' + f')(e' + g')(f' + g').$$



Maximal Independent Set - Example

Multiplying these out and employing the usual identities of Boolean arithmetic, such as

$$aa = a,$$

$$a + a = a,$$

$$a + ab = a,$$

we get

$$\varphi' = b'e'f' + b'e'g' + a'c'd'e'f' + a'c'd'e'g' + b'c'd'f'g'$$

Now if we exclude from the vertex set of G vertices appearing in any one of these five terms, we get a maximal independent set. The five maximal independent sets are

acdf, acdg, bg, bf, and ae

Finding Independence and Chromatic Numbers

Once all the maximal independent sets of G have been obtained, we find the size of the one with the largest number of vertices to get the **independence number** $\beta(G)$.

The independence number of the graph shown earlier is four.

To find the **chromatic number** of G , we must find the minimum number of these (maximal independent) sets, which collectively include all the vertices of G .

For example, in the previous graph, sets $\{a, c, d, f\}$, $\{b, g\}$, and $\{a, e\}$ satisfy this condition. Thus the graph is 3-chromatic.

Chromatic Partitioning

Given a simple, connected graph G , partition all vertices of G into the smallest possible number of disjoint, independent sets.

This problem is known as the **chromatic partitioning** of graphs.

This can be solved by enumerating all maximal independent sets and then selecting the smallest number of sets that include all vertices of the graph.

The following four are some chromatic partitions of the graph shown earlier

$\{(a, c, d, f), (b, g), (e)\},$

$\{(a, c, d, g), (b, f), (e)\},$

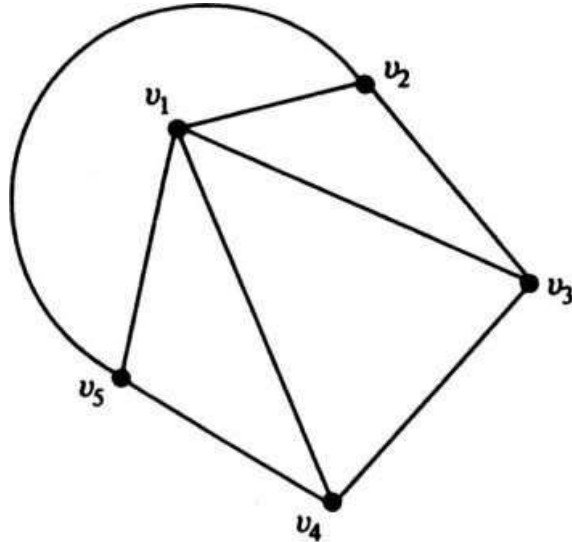
$\{(c, d, f), (b, g), (a, e)\},$

$\{(c, d, g), (b, f), (a, e)\}.$

Uniquely Colorable Graphs

A graph that has only one chromatic partition is called a uniquely colorable graph.

The graph shown earlier is not a uniquely colorable graph, but the one shown here is.



Dominating Sets

A dominating set in a graph G is a set of vertices that dominates every vertex v in G in the following sense: Either v is included in the dominating set or is adjacent to one or more vertices included in the dominating set.

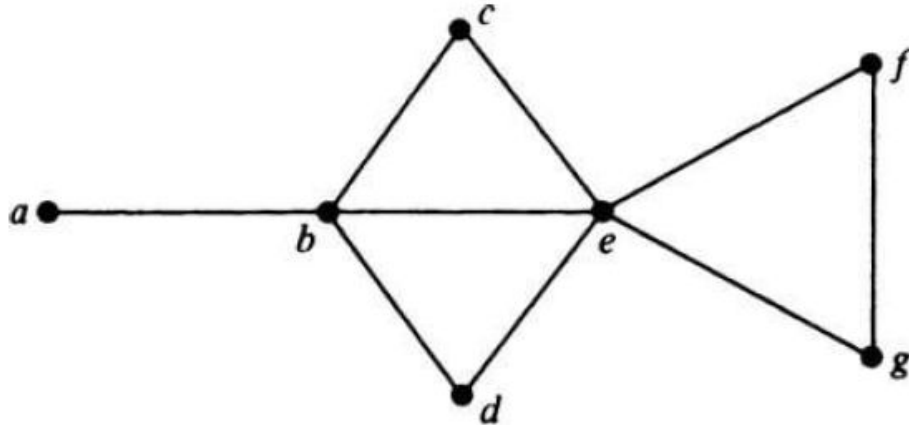
For instance, in the previous graph, the vertex set $\{b, g\}$ is a dominating set.

A dominating set need not be independent. For example, the set of all its vertices is trivially a dominating set in every graph, but it is not necessarily an independent set (unless it is a null graph).

Dominating Sets

Just like we are interested in finding the **maximal independent sets**, similarly we are also interested in finding the **minimal dominating sets**, which are those dominating sets from which no vertex can be removed without destroying the dominance property.

For example, in this graph, $\{b, e\}$ is a minimal dominating set. And so is $\{a, c, d, f\}$.



Dominating Sets

The following observations can be made based on the definition of dominating sets:

1. Any one vertex in a complete graph constitutes a minimal dominating set.
2. Every dominating set contains at least one minimal dominating set.
3. A graph may have many minimal dominating sets, and of different sizes. The number of vertices in the smallest minimal dominating set of a graph G is called the **domination number**, $\alpha(G)$.
4. A minimal dominating set may or may not be independent.

Dominating Sets

The following observations can be made based on the definition of dominating sets:

5. Every maximal independent set is a dominating set. For if an independent set does not dominate the graph, there is at least one vertex that is neither in the set nor adjacent to any vertex in the set. Such a vertex can be added to the independent set without destroying its independence. But then the independent set could not have been maximal.
6. An independent set has the dominance property only if it is a maximal independent set. Thus an independent dominating set is the same as a maximal independent set.
7. In any graph G ,

$$\alpha(G) \leq \beta(G)$$

Finding Minimal Dominating Sets

Just like we found all maximal independent sets of a graph, similarly we can find all minimal dominating sets of the graph using Boolean arithmetic.

To dominate a vertex v_i we must either include v_i or any of the vertices adjacent to v_i . A minimum set satisfying this condition for every vertex v_i is a desired set.

Therefore, for every vertex v_i in G let us form a Boolean product of sums $(v_i + v_{i_1} + v_{i_2} + \dots + v_{i_d})$, where $v_{i_1}, v_{i_2}, \dots, v_{i_d}$ are the vertices adjacent to v_i , and d is the degree of v_i

$$\theta = \prod (v_i + v_{i_1} + v_{i_2} + \dots + v_{i_d}) \text{ for all } v_i \text{ in } G.$$

Finding Minimal Dominating Sets

When θ is expressed as a sum of products, each term in it will represent a minimal dominating set. Let us apply this on the previous graph

$$\theta = (a + b)(b + c + d + e + a)(c + b + e)(d + b + e)(e + b + c + d + f + g)(f + e + g)(g + e + f)$$

Since in Boolean arithmetic $(x + y)x = x$,

$$\begin{aligned}\theta &= (a + b)(b + c + e)(b + d + e)(e + f + g) \\ &= ae + be + bf + bg + acdf + acdg\end{aligned}$$

Each of the six terms in the preceding expression represents a minimal dominating set. And the size of the smallest one is $\alpha(G) = 2$.

CHROMATIC POLYNOMIAL

A given graph G of n vertices can be properly colored in many different ways using a sufficiently large number of colors. This property of a graph is expressed elegantly by means of a polynomial. This polynomial is called the **chromatic polynomial** of G , expressed as $P_n(\lambda)$.

Given λ distinct colors for a graph of n vertices, the value of the chromatic polynomial $P_n(\lambda)$ gives the number of ways of properly coloring the graph, using λ or fewer colors.

CHROMATIC POLYNOMIAL

Let c_i be the different ways of properly coloring G using exactly i different colors. Since i colors can be chosen out of λ colors in ${}^\lambda C_i$ different ways, therefore there are $c_i \times {}^\lambda C_i$ different ways of properly coloring G using exactly i colors out of λ colors.

Since i can be any positive integer from 1 to n (it is not possible to use more than n colors on n vertices), the chromatic polynomial is a sum of these terms:

$$\begin{aligned} P_n(\lambda) &= \sum_{i=1}^n c_i \binom{\lambda}{i} \\ &= c_1 \frac{\lambda}{1!} + c_2 \frac{\lambda(\lambda-1)}{2!} + c_3 \frac{\lambda(\lambda-1)(\lambda-2)}{3!} + \dots \\ &\quad + c_n \frac{\lambda(\lambda-1)(\lambda-2) \dots (\lambda-n+1)}{n!} . \end{aligned}$$

CHROMATIC POLYNOMIAL

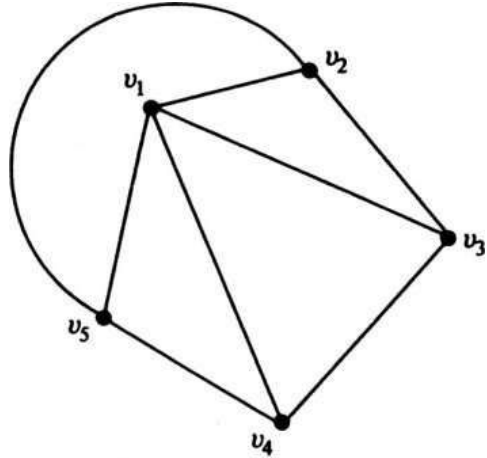
Each c_i has to be evaluated individually for the given graph.

For example, any graph with even one edge requires at least two colors for proper coloring, therefore $c_1 = 0$.

A graph with n vertices and using n different colors can be properly colored in $n!$ ways; that is, $c_n = n!$

CHROMATIC POLYNOMIAL

Let us find the chromatic polynomial of the graph shown here



$$P_5(\lambda) = c_1\lambda + c_2\frac{\lambda(\lambda - 1)}{2} + c_3\frac{\lambda(\lambda - 1)(\lambda - 2)}{3!} \\ + c_4\frac{\lambda(\lambda - 1)(\lambda - 2)(\lambda - 3)}{4!} + c_5\frac{\lambda(\lambda - 1)(\lambda - 2)(\lambda - 3)(\lambda - 4)}{5!}.$$

CHROMATIC POLYNOMIAL

Since the graph has a triangle, it will require at least three different colors for proper coloring. Therefore,

$$c_1 = c_2 = 0 \text{ and } c_5 = 5!$$

Moreover, to evaluate c_3 , suppose that we have three colors x , y , and z . These three colors can be assigned properly to vertices v_1 , v_2 , and v_3 in $3! = 6$ different ways. Having done that, we have no more choices left, because vertex v_5 must have the same color as v_3 , and v_4 must have the same color as v_2 . Therefore,

$$c_3 = 6$$

CHROMATIC POLYNOMIAL

Similarly, with four colors, v_1, v_2 , and v_3 can be properly colored in $4 \cdot 6 = 24$ different ways. The fourth color can be assigned to v_4 or v_5 , thus providing two choices. The fifth vertex provides no additional choice. Therefore,

$$c_4 = 24 \cdot 2 = 48$$

Substituting these coefficients in $P_5(\lambda)$, we get,

$$\begin{aligned} P_5(\lambda) &= \lambda(\lambda - 1)(\lambda - 2) + 2\lambda(\lambda - 1)(\lambda - 2)(\lambda - 3) + \lambda(\lambda - 1)(\lambda - 2)(\lambda - 3)(\lambda - 4) \\ &= \lambda(\lambda - 1)(\lambda - 2)(\lambda^2 - 5\lambda + 7) \end{aligned}$$

The presence of factors $\lambda - 1$ and $\lambda - 2$ indicates that G is at least 3-chromatic.

THEOREM 8-4 and THEOREM 8-5

THEOREM 8-4: *“A graph of n vertices is a complete graph if and only if its chromatic polynomial is $P_n(\lambda) = \lambda(\lambda - 1)(\lambda - 2) \dots (\lambda - n + 1)$ ”*

THEOREM 8-5: *“An n -vertex graph is a tree if and only if its chromatic polynomial is $P_n(\lambda) = \lambda(\lambda - 1)^{n-1}$ ”*

THEOREM 8-6

Let a and b be two nonadjacent vertices in a graph G . Let G' be a graph obtained by adding an edge between a and b . Let G'' be a simple graph obtained from G by fusing the vertices a and b together and replacing sets of parallel edges with single edges. Then

$$P_n(\lambda) \text{ of } G = P_n(\lambda) \text{ of } G' + P_{n-1}(\lambda) \text{ of } G''$$

Proof: The number of ways of properly coloring G can be grouped into two cases, one such that vertices a and b are of the same color and the other such that a and b are of different colors.

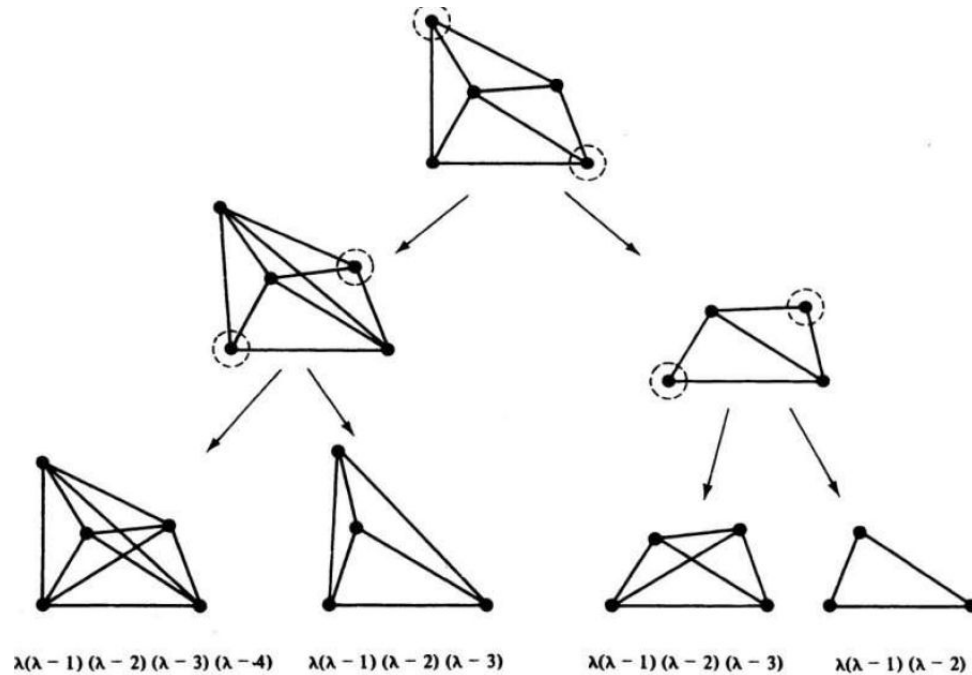
THEOREM 8-6

The number of ways of properly coloring G such that a and b have different colors
= number of ways of properly coloring G'

Also, the number of ways of properly coloring G such that a and b have the same color = number of ways of properly coloring G'' , so

$$P(\lambda) \text{ of } G = P_n(\lambda) \text{ of } G' + P_{n-1}(\lambda) \text{ of } G''$$

THEOREM 8-6



$$\begin{aligned}
 P_5(\lambda) \text{ of } G &= \lambda(\lambda-1)(\lambda-2) + 2\lambda(\lambda-1)(\lambda-2)(\lambda-3) \\
 &\quad + \lambda(\lambda-1)(\lambda-2)(\lambda-3)(\lambda-4) \\
 &= \lambda(\lambda-1)(\lambda-2)(\lambda^2 - 5\lambda + 7)
 \end{aligned}$$

Edge Coloring

Until now, we have been concerned with proper coloring of the “**vertices**” in a graph. Suppose that we are interested in coloring the “**edges**” rather than the vertices.

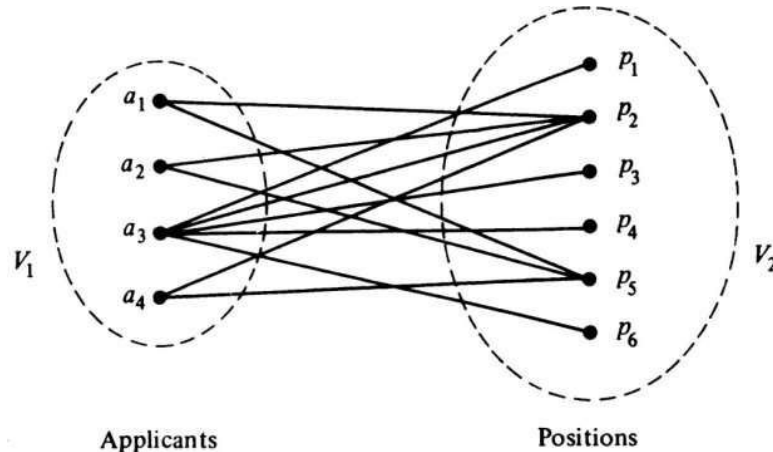
It is reasonable to call two edges **adjacent** if they have one end vertex in common (but are not parallel). A proper coloring of edges then requires that adjacent edges should be of different colors.

Moreover, a set of edges in which no two are adjacent is similar to an independent set of vertices. Such a set of edges is called a **matching**.

MATCHINGS

Suppose that four applicants a_1 , a_2 , a_3 , and a_4 are available to fill six vacant positions p_1 , p_2 , p_3 , p_4 , p_5 , and p_6 . Applicant a_1 is qualified to fill position p_2 or p_5 . Applicant a_2 can fill p_2 or p_5 . Applicant a_3 is qualified for p_1 , p_2 , p_3 , p_4 , or p_6 . Applicant a_4 can fill jobs p_2 or p_5 .

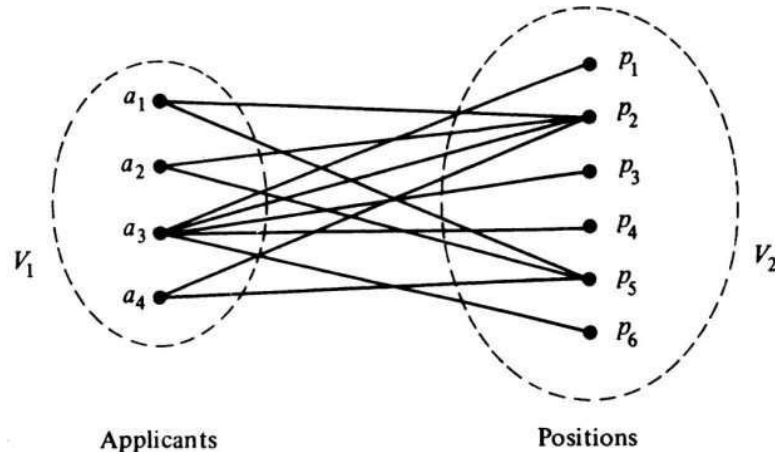
This situation is represented by the graph shown here



MATCHINGS

The vacant positions and applicants are represented by vertices. The edges represent the qualifications of each applicant for filling different positions.

The graph clearly is bipartite, as the vertices fall into two sets $V_1 = \{a_1, a_2, a_3, a_4\}$ and $V_2 = \{p_1, p_2, p_3, p_4, p_5, p_6\}$.



MATCHINGS

The question we are trying to answer here is:

Is it possible to hire all the applicants and assign each a position for which he is suitable? If the answer is no, what is the maximum number of positions that can be filled from the given set of applicants?

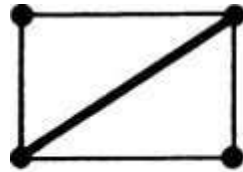
This is a problem of matching (or assignment) of one set of vertices into another. More formally, a matching in a graph is a subset of edges in which no two edges are adjacent.

A single edge in a graph is obviously a valid matching. But we are usually interested in finding the **maximal matching**.

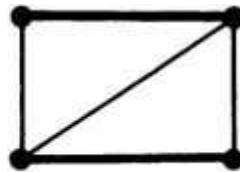
MATCHINGS

A maximal matching is a matching to which no edge in the graph can be added. For example, in a complete graph of three vertices (i.e., a triangle) any single edge is a maximal matching.

The edges shown by heavy lines in this figure are two maximal matchings of this graph.



(a)



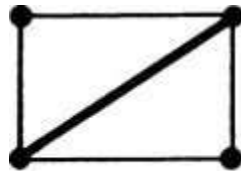
(b)

MATCHINGS

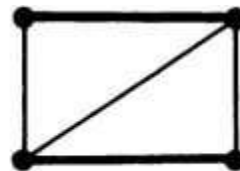
Clearly, a graph may have many different maximal matchings, and of different sizes. Among these, the maximal matchings with the largest number of edges are called the **largest maximal matching**.

In the right subfigure, a largest maximal matching is shown in heavy lines. The left one is a maximal matching, but not the largest one.

The number of edges in a largest maximal matching is called the matching number of the graph. For this graph, it is 2.



(a)



(b)

MATCHINGS

Although matching is defined for any graph, it is mostly studied in the context of bipartite graphs, as suggested by the introduction to this section.

In a bipartite graph having a vertex partition V_1 and V_2 , a **complete matching** of vertices in set V_1 into those in V_2 is a matching in which there is one edge incident with every vertex in V_1 .

In other words, every vertex in V_1 is matched against some vertex in V_2 .

MATCHINGS

For the existence of a complete matching of set V_1 into set V_2 , first we must have at least as many vertices in V_2 as there are in V_1 .

In other words, there must be at least as many vacant positions as the number of applicants if all the applicants are to be hired.

This condition, however, is not sufficient. For example, in the bipartite graph figure shown earlier, although there are six positions and four applicants (more positions available than the number of applicants), still a complete matching does not exist. Of the three applicants a_1 , a_2 and a_4 , each qualifies for the same two positions p_2 and p_5 , and therefore one of the three applicants cannot be matched.

MATCHINGS

This leads us to another necessary condition for a complete matching:

Every subset of r vertices in V_1 must collectively be adjacent to at least r vertices in V_2 , for all values of $r = 1, 2, \dots, |V_1|$.

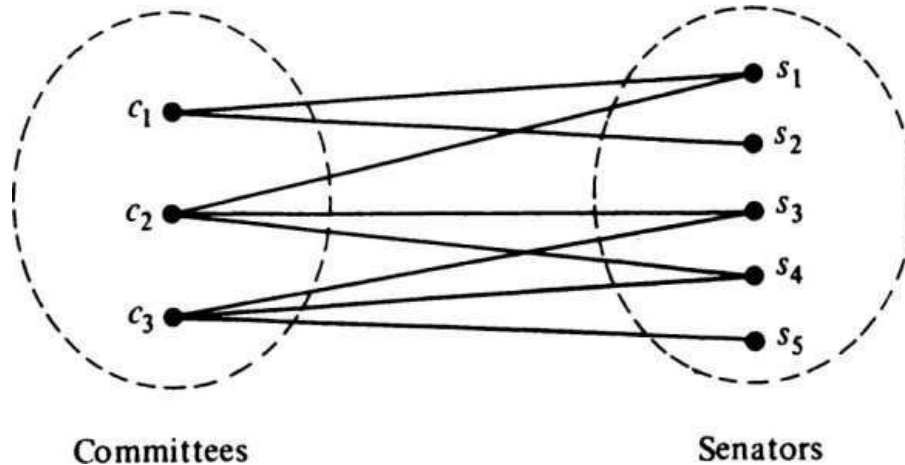
A surprising fact is that this condition is also sufficient for the existence of a complete matching. This is outlined in Theorem 8-7.

THEOREM 8-7: *“A complete matching of V_1 into V_2 in a bipartite graph exists if and only if every subset of r vertices in V_1 is collectively adjacent to r or more vertices in V_2 for all values of r .”*

MATCHINGS

Let us illustrate this important theorem with an example.

Problem of Distinct Representatives: Five senators $s_1, s_2, s_3, s_4,$ and s_5 are members of three committees, $c_1, c_2,$ and c_3 . The membership is shown in the following figure.

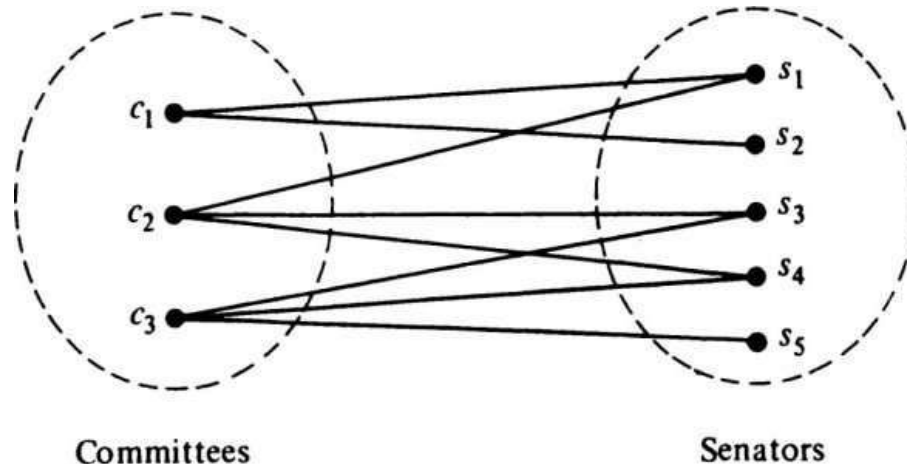


MATCHINGS

One member from each committee is to be represented in a super-committee.

Is it possible to send one distinct representative from each of the committees?

Let us use **Theorem 8-7** and check if r vertices from V_1 are collectively adjacent to at least r vertices from V_2 , for all values of r .



MATCHINGS

The result of applying **THEOREM 8-7** is shown in the following Table (ignore the last column for the time being).

r	V_1	V_2	r-q
1	$\{c_1\}$	$\{s_1, s_2\}$	-1
	$\{c_2\}$	$\{s_1, s_3, s_4\}$	-2
	$\{c_3\}$	$\{s_3, s_4, s_5\}$	-2
2	$\{c_1, c_2\}$	$\{s_1, s_2, s_3, s_4\}$	-2
	$\{c_2, c_3\}$	$\{s_1, s_3, s_4, s_5\}$	-2
	$\{c_1, c_3\}$	$\{s_1, s_2, s_3, s_4, s_5\}$	-3
3	$\{c_1, c_2, c_3\}$	$\{s_1, s_2, s_3, s_4, s_5\}$	-2

MATCHINGS

The problem of distinct representatives just solved was a small one. A larger problem would have become unwieldy. If there are M vertices in V_1 , **Theorem 8-7** requires that we take all $2^M - 1$ nonempty subsets of V_1 and find the number of vertices of V_2 adjacent collectively to each of these.

Theorem 8-8 gives us a simplification of **Theorem 8-7**, which will suffice for detection of a complete matching in any large graph in most cases.

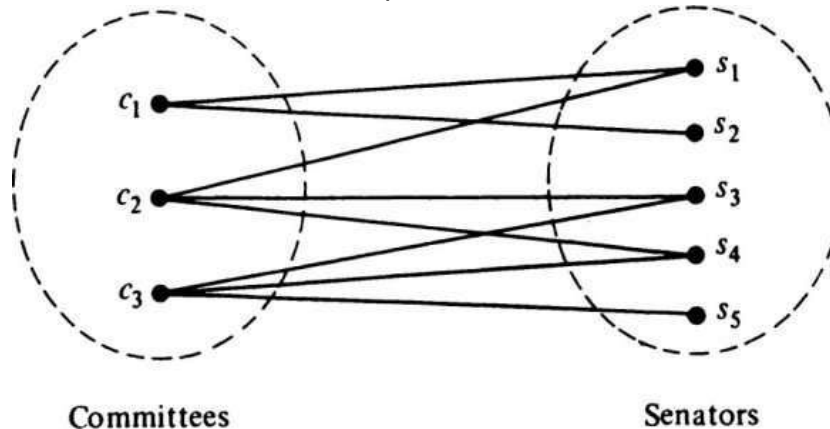
THEOREM 8-8: In a bipartite graph a complete matching of V_1 into V_2 exists if (but not only if) there is a positive integer m for which the following condition is satisfied:

$$\text{degree of every vertex in } V_1 \geq m \geq \text{degree of every vertex in } V_2$$

MATCHINGS

Applying THEOREM 8-8 on this graph, we get

degree of every vertex in $V_1 \geq 2 \geq$ degree of every vertex in V_2



Therefore, there exists a complete matching.

For the previous bipartite graph (applicants and positions), no such number is found, because the degree of $p_2 = 4 >$ degree of a_1 .

Maximal Matching

We have seen how to find whether a complete matching exists or not. Now we are interested in finding a largest maximal matching, that is, to pair off as many vertices of V_1 with those in V_2 as possible, even if a complete matching does not exist.

For this purpose, let us define a new term called **deficiency**, denoted as $\delta(G)$, of a bipartite graph G .

A set of r vertices in V_1 is collectively incident on, say, q vertices of V_2 . Then the maximum value of the number $r - q$ taken over all values of $r = 1, 2, \dots$ and all subsets of V_1 is called the deficiency $\delta(G)$ of the bipartite graph G .

Maximal Matching

Theorem 8-7, expressed in terms of the deficiency, states that a complete matching in a bipartite graph G exists if and only if $\delta(G) \leq 0$.

For example, the deficiency of the bipartite graph shown earlier (Committee-Senator graph) was -1 (the largest number in the last column of the Table). For the Applicants-Positions graph, you can verify that the deficiency was $+1$.

THEOREM 8-9: The maximal number of vertices in set V_1 that can be matched into V_2 is equal to

$$\text{number of vertices in } V_1 - \delta(G)$$

Matching and Adjacency Matrix

Consider a bipartite graph G with nonadjacent sets of vertices V_1 and V_2 , having number of vertices n_1 and n_2 , respectively, and let $n_1 \leq n_2$, $n_1 + n_2 = n$, the number of vertices in G . The adjacency matrix $X(G)$ of G can be written in the form

$$X(G) = \begin{bmatrix} 0 & X_{12} \\ X_{12}^T & 0 \end{bmatrix}$$

where the submatrix X_{12} is the n_1 by n_2 , $(0, 1)$ -matrix containing the information as to which of the n_1 vertices of V_1 are connected to which of the n_2 vertices of V_2 .

Matching and Adjacency Matrix

A matching V_1 into V_2 corresponds to a selection of the 1's in the matrix X_{12} such that no line (i.e., a row or a column) has more than one 1.

The matching is complete if the n_1 by n_2 matrix made of selected 1's has exactly one 1 in every row.

For example, the X_{12} matrix for the Committee-Senator graph is

$$X_{12} = \begin{matrix} & \begin{matrix} s_1 & s_2 & s_3 & s_4 & s_5 \end{matrix} \\ \begin{matrix} c_1 \\ c_2 \\ c_3 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \end{matrix},$$

$$n_1 = 3, \quad n_2 = 5, \quad n = 8, \quad \text{and} \quad n_1 \leq n_2,$$

$$V_1 = \{c_1, c_2, c_3\}$$

$$V_2 = \{s_1, s_2, s_3, s_4, s_5\}.$$

Matching and Adjacency Matrix

A complete matching of V_1 into V_2 is given by

$$\mathbf{M} = \begin{matrix} & \begin{matrix} s_1 & s_2 & s_3 & s_4 & s_5 \end{matrix} \\ \begin{matrix} c_1 \\ c_2 \\ c_3 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

A maximal matching corresponds to the selection of a largest possible number of 1's from X_{12} such that no row in it has more than one 1.

COVERINGS

In a graph G , a set $\mathbf{g}$ of edges is said to cover $\mathbf{G}$ if every vertex in $\mathbf{G}$ is incident on at least one edge in $\mathbf{g}$. A set of edges that covers a graph G is said to be an edge covering, a covering subgraph, or simply a covering of G .

Examples of coverings:

- A graph G is trivially its own covering
- A spanning tree in a connected graph (or a spanning forest in an unconnected graph) is another covering
- A Hamiltonian circuit (if it exists) in a graph is also a covering.

COVERINGS

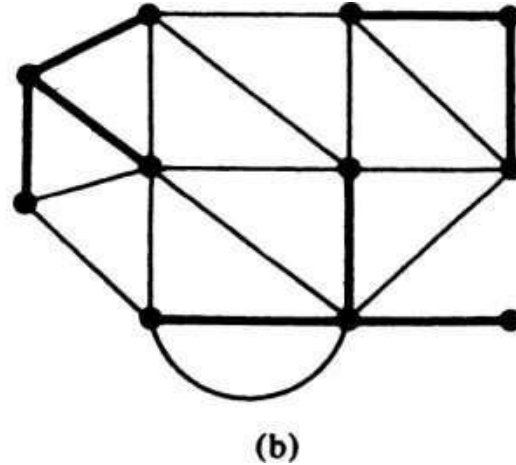
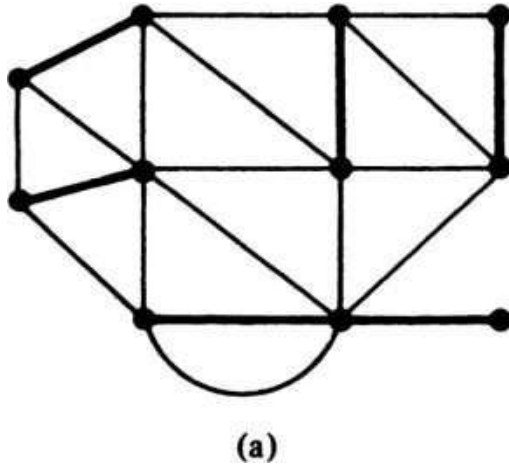
We are not interested in just any covering.

We have already dealt with some coverings with specific properties, such as spanning trees and Hamiltonian circuits.

In this section we shall investigate the **minimal covering**, which is a covering from which no edge can be removed without destroying its ability to cover the graph.

COVERINGS

The figure below shows a graph and two of its minimal coverings (in heavy lines).



COVERINGS

We can make the following observations:

1. A covering exists for a graph if and only if the graph has no isolated vertex.
2. A covering of an n -vertex graph will have at least $\lceil n/2 \rceil$ edges. ($\lceil x \rceil$ denotes the smallest integer not less than x .)
3. Every pendant edge in a graph is included in every covering of the graph.
4. Every covering contains a minimal covering.
5. If we denote the remaining edges of a graph by $(G - g)$, the set of edges g is a covering if and only if, for every vertex v , the degree of vertex v in $(G - g) \leq (\text{degree of vertex } v \text{ in } G) - 1$.

COVERINGS

We can make the following observations:

6. No minimal covering can contain a circuit, for we can always remove an edge from a circuit without leaving any of the vertices in the circuit uncovered.
 - a. Therefore, a minimal covering of an n -vertex graph can contain no more than $n - 1$ edges.
7. A graph, in general, has many minimal coverings, and they may be of different sizes (i.e., consisting of different numbers of edges). The number of edges in a minimal covering of the smallest size is called the **covering number** of the graph.

THEOREM 8-10

“A covering g of a graph is minimal if and only if g contains no paths of length three or more.”

Proof: Suppose that a covering g contains a path of length three, and it is

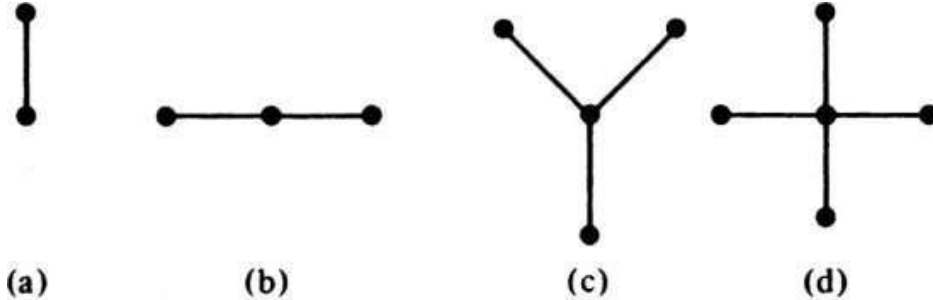
$$v_1 e_1 v_2 e_2 v_3 e_3 v_4$$

It is obvious that edge e_2 can be removed without leaving its end vertices v_2 and v_3 uncovered.

Therefore, g is not a minimal covering.

THEOREM 8-10

Conversely, if a covering g contains no path of length three or more, all its components must be **star graphs**, as shown here.

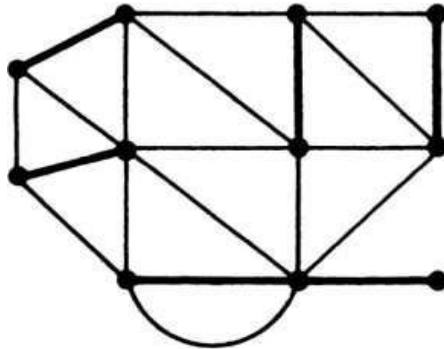


From a star graph no edge can be removed without leaving a vertex uncovered. That is, g must be a minimal covering.

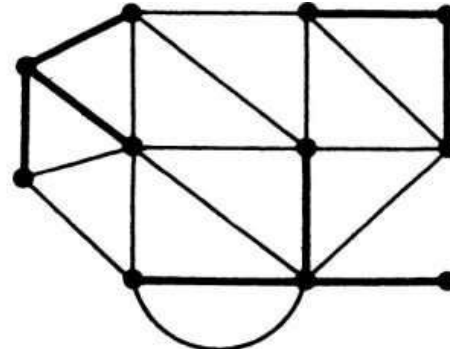
Smallest Minimal Covering

Suppose that the graph we saw earlier represents the street map of a part of a city. Each of the vertices is a potential trouble spot and must be kept under the surveillance of a patrol car.

How will you assign a minimum number of patrol cars to keep every vertex covered?



(a)

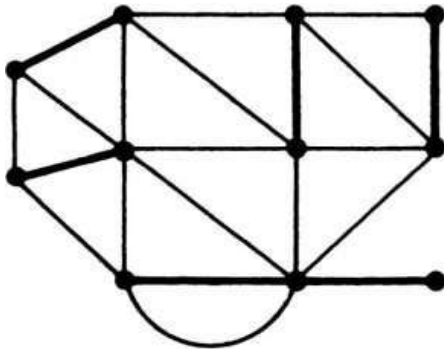


(b)

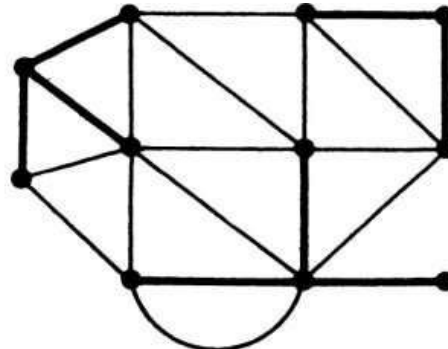
Smallest Minimal Covering

The answer is to find a smallest minimal covering. Here, the covering on the left requires 6 patrol cars (6 edges), and the right one requires 8 cars, although both are minimal coverings.

Clearly, since there are 11 vertices and no edge can cover more than two, less than six edges cannot cover the graph.



(a)



(b)

Minimization of Switching Functions

An important step in the logical design of a digital machine is to minimize Boolean functions before implementing them.

Suppose we are interested in building a logical circuit that gives the following function F of four Boolean variables w , x , y , and z .

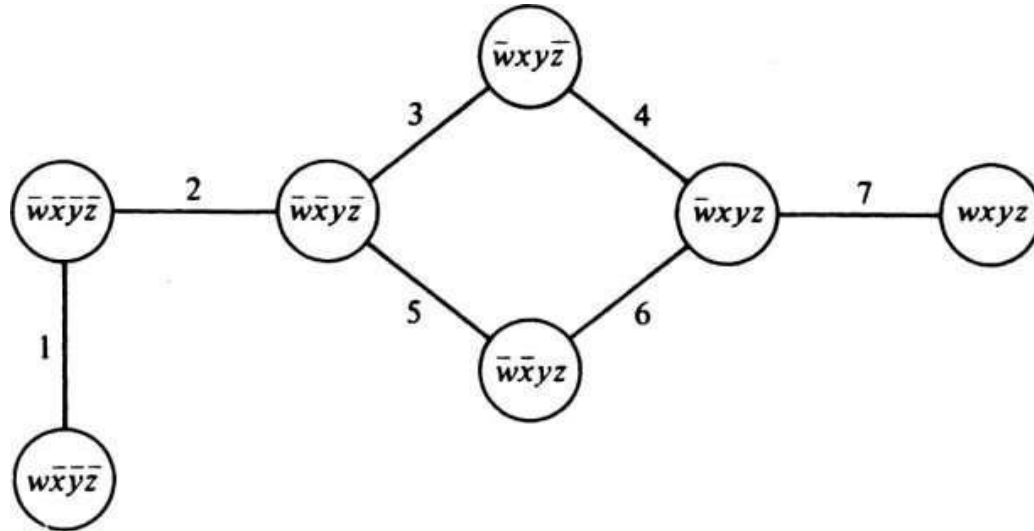
$$F = \bar{w}\bar{x}\bar{y}\bar{z} + \bar{w}\bar{x}y\bar{z} + w\bar{x}\bar{y}\bar{z} + \bar{w}\bar{x}yz + \bar{w}xy\bar{z} + \bar{w}xyz + wxyz,$$

Let us represent each of the seven terms in F by a vertex, and join every pair of vertices that differ only in one variable. An edge between two vertices represents a term with three variables.

A minimal cover of this graph will represent a simplified form of F , performing the same function as F , but with less logic hardware.

Minimization of Switching Functions

The pendant edges 1 and 7 must be included in every covering of the graph. Two additional edges 3 and 6 (or 4 and 5 or 3 and 5) will cover the remainder.

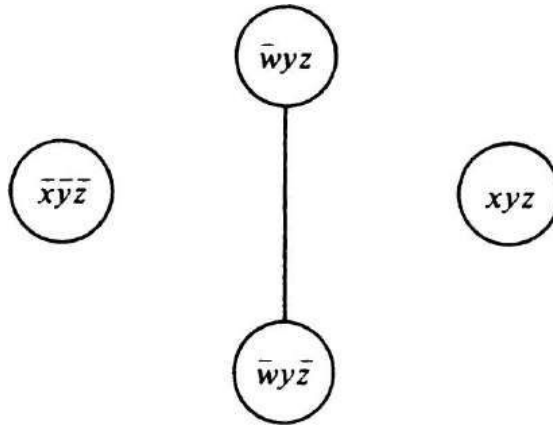


Minimization of Switching Functions

Thus a simplified version of F is

$$F = \bar{x}\bar{y}\bar{z} + xyz + \bar{w}y\bar{z} + \bar{w}yz.$$

This expression can again be represented by a graph of four vertices:



Minimization of Switching Functions

Applying the same process on the remaining graph, we see that one edge will cover the remaining two vertices in that graph.

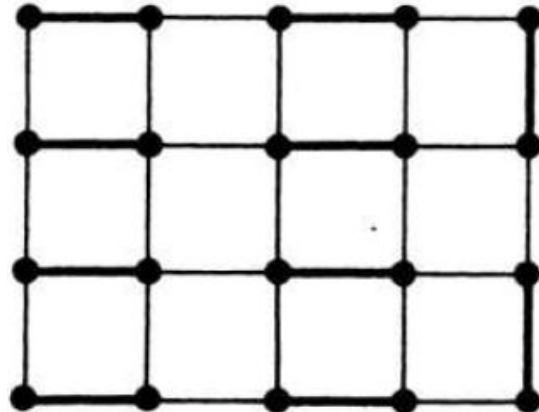
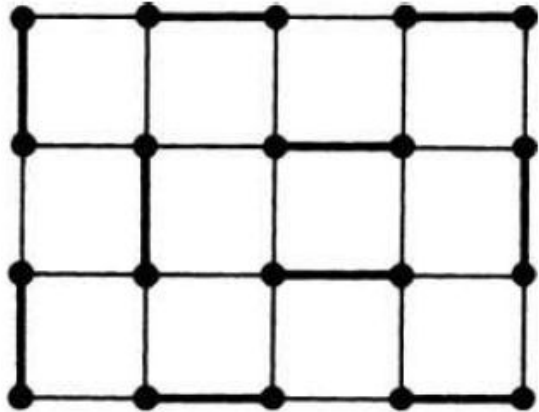
Thus the minimized Boolean expression is

$$F = \bar{x}\bar{y}\bar{z} + xyz + \bar{w}y.$$

Dimer Problem

Those who study crystal physics usually face the dimer problem, which can actually be converted into a graph theory problem.

In the dimer problem, one is required to find the number of ways in which all atoms on a two-dimensional lattice can be paired off as molecules (each consisting of two atoms).



Dimer Problem

The problem is equivalent to finding all different coverings of a given graph such that every vertex in the covering is of degree one. Such a covering in which every vertex is of degree one is called a **dimer covering** or a **1-factor**.

A dimer covering is obviously a matching because no two edges in it are adjacent. Moreover, a dimer covering is a maximal matching.

A graph must have an even number of vertices to have a dimer covering. This is a necessary condition, but not a sufficient one.

FOUR-COLOR PROBLEM

So far we have considered proper coloring of vertices and proper coloring of edges. Now, we will consider the **proper coloring of regions** in a planar graph.

The regions of a planar graph are said to be properly colored if no two contiguous or adjacent regions have the same color. Two regions are said to be adjacent if they have a common edge between them.

We are interested in a coloring that uses the minimum number of colors. This leads us to the most famous conjecture in graph theory. The conjecture is that every map (i.e., a planar graph) can be properly colored with four colors.

Vertex Coloring vs Region Coloring

From Chapter 5 we know that a graph has a dual if and only if it is planar. Therefore, coloring the regions of a planar graph G is equivalent to coloring the vertices of its dual G^* , and vice versa.

Thus the four-color conjecture can be restated as follows: ***“Every planar graph has a chromatic number of four or less.”***

It is quite difficult to prove that four colors is enough for all possible planar graphs, but proving that five colors is enough is fairly easy. This is proved in **Theorem 8-11**.

THEOREM 8-11

“The vertices of every planar graph can be properly colored with five colors.”

Proof: This theorem can be proved by induction.

The theorem is trivially true for all graphs of five or less vertices.

As our hypothesis, let us assume that vertices of every planar graph with $n - 1$ vertices can be properly colored with five colors.

Then, if we prove that any planar graph G with n vertices will require no more than five colors, we shall have proved the theorem.

THEOREM 8-11

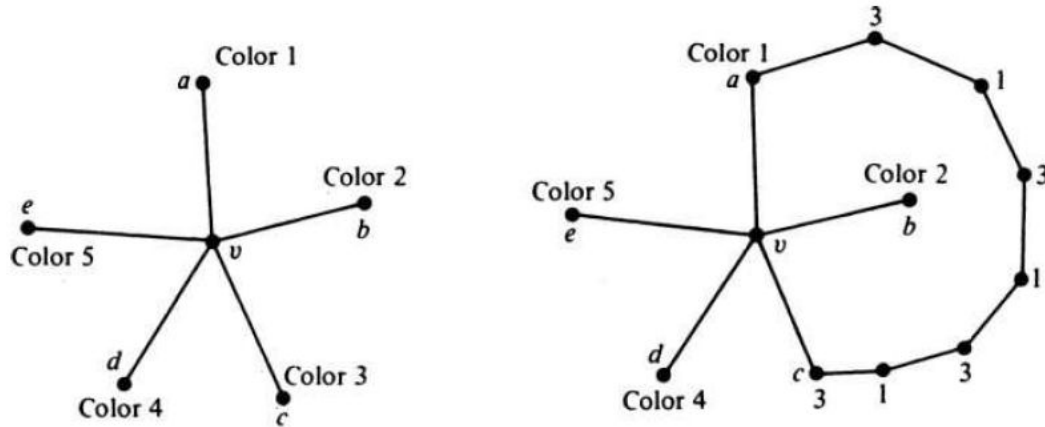
Since G is planar, it must have at least one vertex with degree five or less. Let this vertex be v .

Let G' be a graph (of $n - 1$ vertices) obtained from G by deleting vertex v . Graph G' requires no more than five colors, according to the induction hypothesis.

Suppose that the vertices in G' have been properly colored, and now we add back to it the vertex v and all edges incident on v . If the degree of v is 1, 2, 3, or 4, we have no difficulty in assigning a proper color to v .

THEOREM 8-11

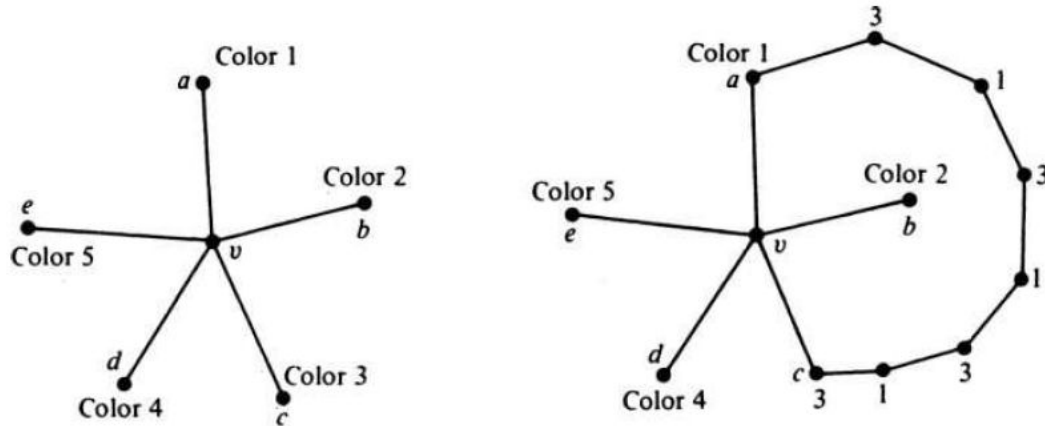
This leaves only the case in which the degree of v is five, and all the five colors have been used in coloring the vertices adjacent to v , as shown below.



Suppose that there is a path in G' between vertices a and c colored alternately with colors 1 and 3, as shown in the right subfigure.

THEOREM 8-11

Then a similar path between b and d , colored alternately with colors 2 and 4, cannot exist; otherwise, these two paths will intersect and cause G to be nonplanar.



THEOREM 8-11

If there is no path between b and d colored alternately with colors 2 and 4, starting from vertex b we can interchange colors 2 and 4 of all vertices connected to b through vertices of alternating colors 2 and 4.

This interchange will paint vertex b with color 4 and yet keep G' properly colored. Since vertex d is still with color 4, we have color 2 left over with which to paint vertex v .

Regularization of a Planar Graph

Removing every vertex of degree one (together with the pendant edge) from the graph G does not affect the regions of a planar graph. Nor does the elimination of every vertex of degree two, by merging the two edges in series.

Now consider a typical vertex v of degree four or more in a planar graph. Let us replace vertex v by a small circle with as many vertices as there were incidences on v . This results in a number of vertices each of degree three.

Performing this transformation on every vertex of degree four or more in a planar graph G will produce another planar graph H in which every vertex is of degree three. Such a transformation may be called **regularization of a planar graph**.

Regularization of a Planar Graph

When the regions of H have been properly colored, a proper coloring of the regions of G can be obtained simply by shrinking each of the new regions back to the original vertex.

Clearly, if H can be colored with four colors, so can G . Thus, for map-coloring problems, it is sufficient to confine oneself to (connected) planar, regular graphs of degree three.

